WORKSHEET · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 2: Analysing Data: Average, Maximum, Minimum

Module 1 | 45 minutes | Data algorithms | Grades 6–8 · ages 11–14

Name: Class: Date:

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. mean (average) ___ A. Checking each item once, in order.
b. maximum / minimum ___ B. The largest / smallest value.
c. linear scan ___ C. Total divided by the number of items.

2. Plan your code

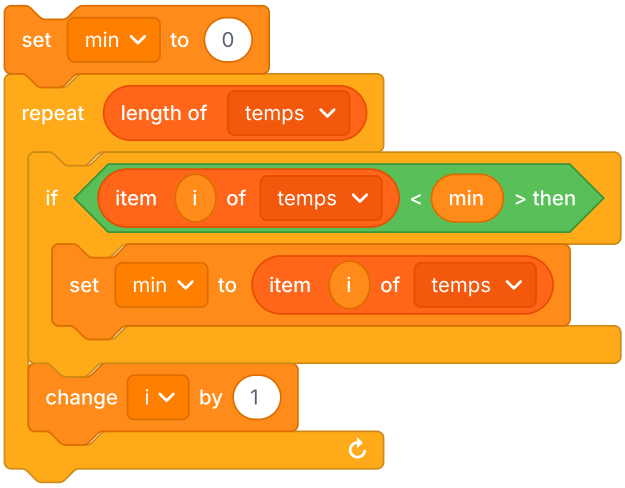
Before you build it: Maximum: find the highest temperature. Set max to item 1 of temps; traverse the list; if item i > max, set max to item i. Say the max (27). Write or draw the blocks you will use.

3. Plan your code

Before you build it: Average: total the list, then set average to total / length of temps and say it. The average of 2, 4, 6, 8 is 5. Write or draw the blocks you will use.

4. Debug the minimum

This minimum finder says 0, but the smallest temperature is 18. What's the bug? Circle one.



a. A. min should start as item 1, not 0
b. B. The < should be >
c. C. i should change before the if

5. Explain your thinking

You have a list of your class's test scores. Describe two questions you could answer with average, maximum or minimum, and why they'd be useful.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can calculate an average.
☐ I can find the max and min with a scan.
☐ I can explain why initialisation matters.